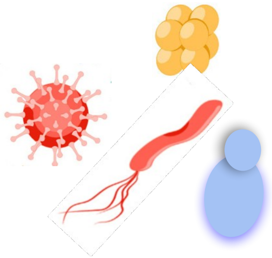
Introduction to mNGS*& lab setup

*metagenomic Next-Generation Sequencing

Introduction



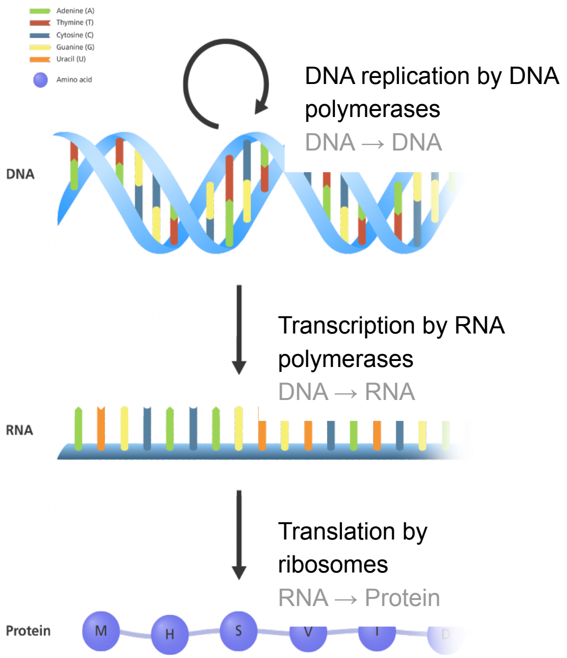
Pathogen landscape

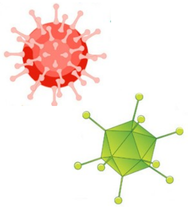
Sample landscape

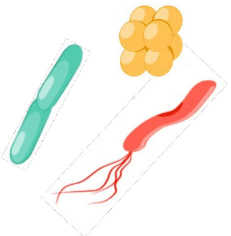
NGS toolbox for
infectious diseases

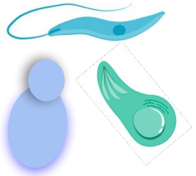
Pathogen landscape

Organisms vary widely in their genetic makeup









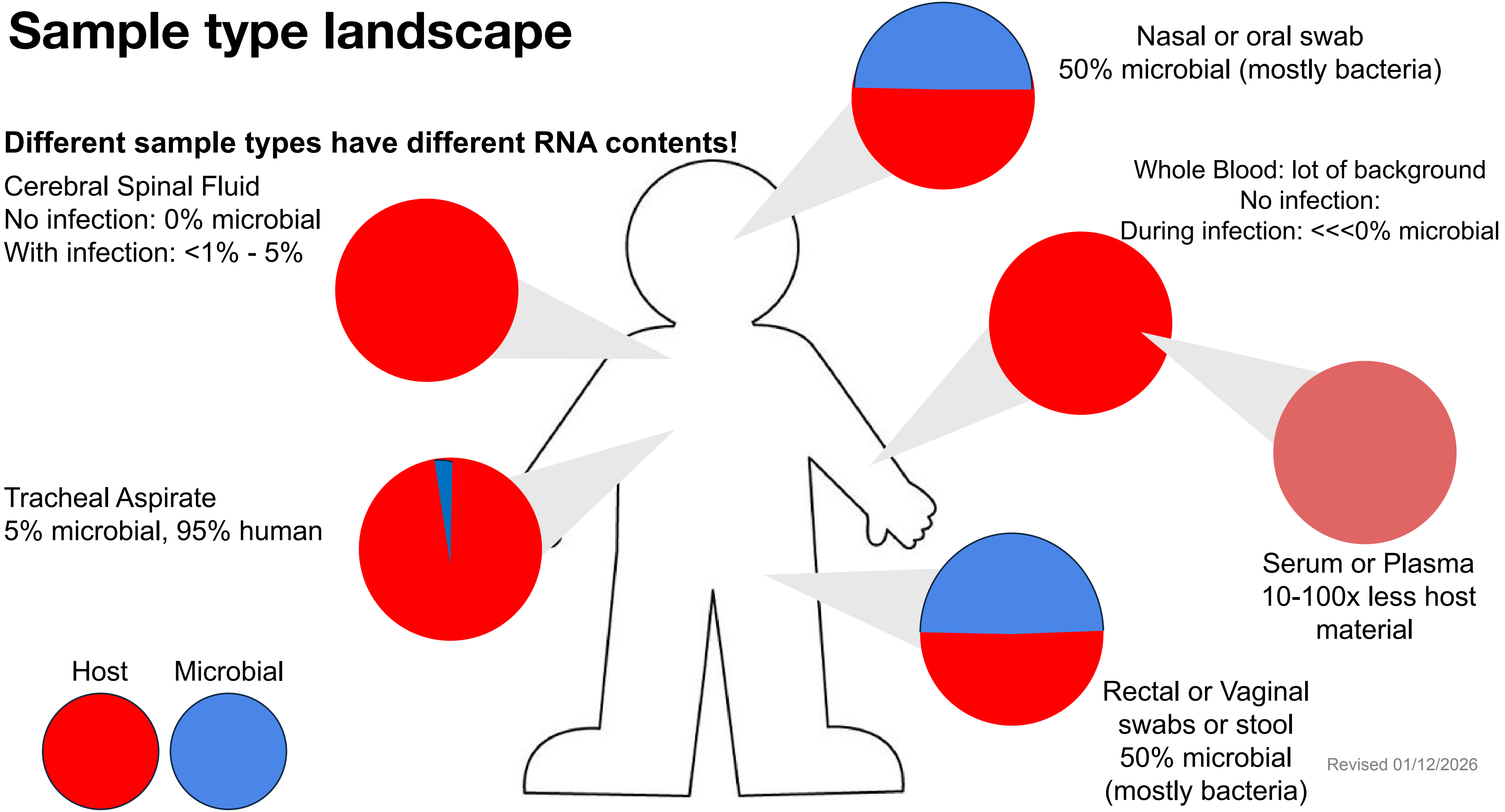
	Viruses	Bacteria	Eukaryotic microbes (yeast, amoebas, worms, etc)
Genome composition	ssRNA, ssDNA, dsDNA, or dsRNA	dsDNA	dsDNA
Ribosomal RNA	None, uses host ribosomes	16S rRNA	18S rRNA
Genome size	2 Kb - 1 Mb	2 Mb - 8 Mb	200 Kb - 100,000 Mb

Adapted from yourgenome.org

Revised 01/12/2026

Sample type landscape

Different sample types have different RNA contents!



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NGS toolbox for infectious diseases

	Whole genome & transcriptome sequencing
Starting material	DNA/RNA from (cultured) isolate
Primers used to amplify sequences	Random primers
What is sequenced?	Organism of interest
Resulting data	DNA sequence or gene expression of target organism

NGS toolbox for infectious diseases

	Whole genome & transcriptome sequencing	Amplicon sequencing
Starting material	DNA/RNA from (cultured) isolate	DNA/RNA from a sample
Primers used to amplify sequences	Random primers	Organism-specific primers
What is sequenced?	Organism of interest	Organism of interest
Resulting data	DNA sequence or gene expression of target organism	RNA/DNA sequence of targeted organism

NGS toolbox for infectious diseases

	Whole genome & transcriptome sequencing	Amplicon sequencing	16S & 18S rRNA sequencing
Starting material	DNA/RNA from (cultured) isolate	DNA/RNA from a sample	DNA/RNA from a sample
Primers used to amplify sequences	Random primers	Organism-specific primers	16S/18S-specific primers
What is sequenced?	Organism of interest	Organism of interest	Only rRNA from bacteria, fungi and eukaryotic pathogens in sample
Resulting data	DNA sequence or gene expression of target organism	RNA/DNA sequence of targeted organism	Only rRNA sequences of bacteria, fungi and eukaryotic pathogens

NGS toolbox for infectious diseases

	Whole genome & transcriptome sequencing	Amplicon sequencing	16S & 18S rRNA sequencing	Metagenomic (shotgun) sequencing
Starting material	DNA/RNA from (cultured) isolate	DNA/RNA from a sample	DNA/RNA from a sample	DNA/RNA from a sample
Primers used to amplify sequences	Random primers	Organism-specific primers	16S/18S-specific primers	Random primers
What is sequenced?	Organism of interest	Organism of interest	Only rRNA from bacteria, fungi and eukaryotic pathogens in sample	DNA/RNA from all organisms in sample
Resulting data	DNA sequence or gene expression of target organism	RNA/DNA sequence of targeted organism	Only rRNA sequences of bacteria, fungi and eukaryotic pathogens	DNA/RNA sequences from all organisms in sample

NGS toolbox for infectious diseases

	Whole genome & transcriptome sequencing	Amplicon sequencing	16S & 18S rRNA sequencing	Metagenomic (shotgun) sequencing
Starting material	DNA/RNA from (cultured) isolate	DNA/RNA from a sample	DNA/RNA from a sample	DNA/RNA from a sample
Primers used to amplify sequences	Random primers	Organism-specific primers	16S/18S-specific primers	Random primers
What is sequenced?	Organism of interest	Organism of interest	Only rRNA from bacteria, fungi and eukaryotic pathogens in sample	DNA/RNA from all organisms in sample
Resulting data	DNA sequence or gene expression of target organism	RNA/DNA sequence of targeted organism	Only rRNA sequences of bacteria, fungi and eukaryotic pathogens	DNA/RNA sequences from all organisms in sample
Use cases	Fundamental research on cultured organisms	Full genomes from clinical samples for genomic epidemiology	Exploratory studies on environmental samples	Outbreak research, co-infections, ...

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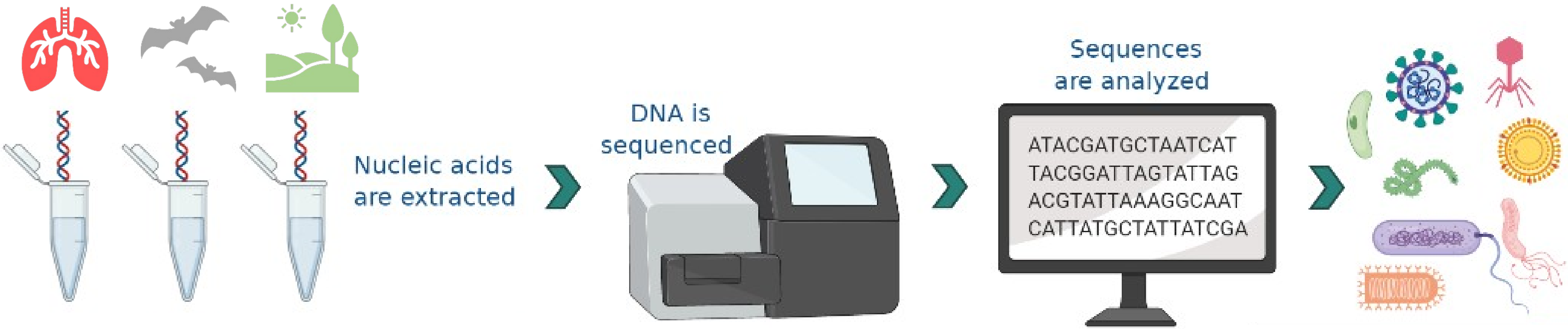
mNGS



Metagenomic next generation sequencing

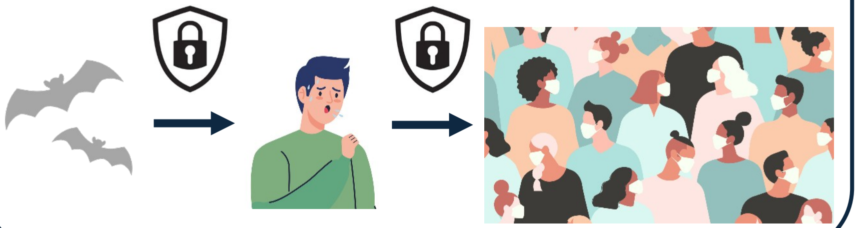
Comprehensive analysis of <u>all of the genetic material</u> present in a sample

RNA or DNA mNGS? That depends on your pathogen(s) of interest...

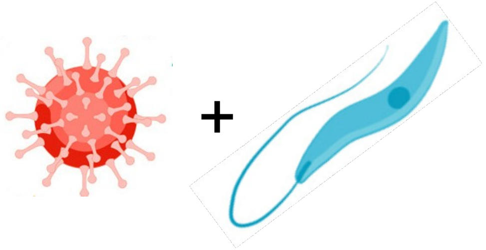
Use cases for mNGS

Identify a novel pathogen

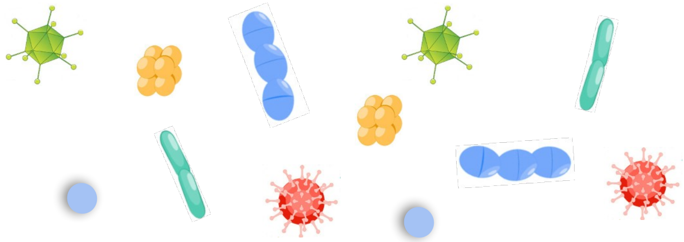
Outbreak detection and prevention



Identify coinfections

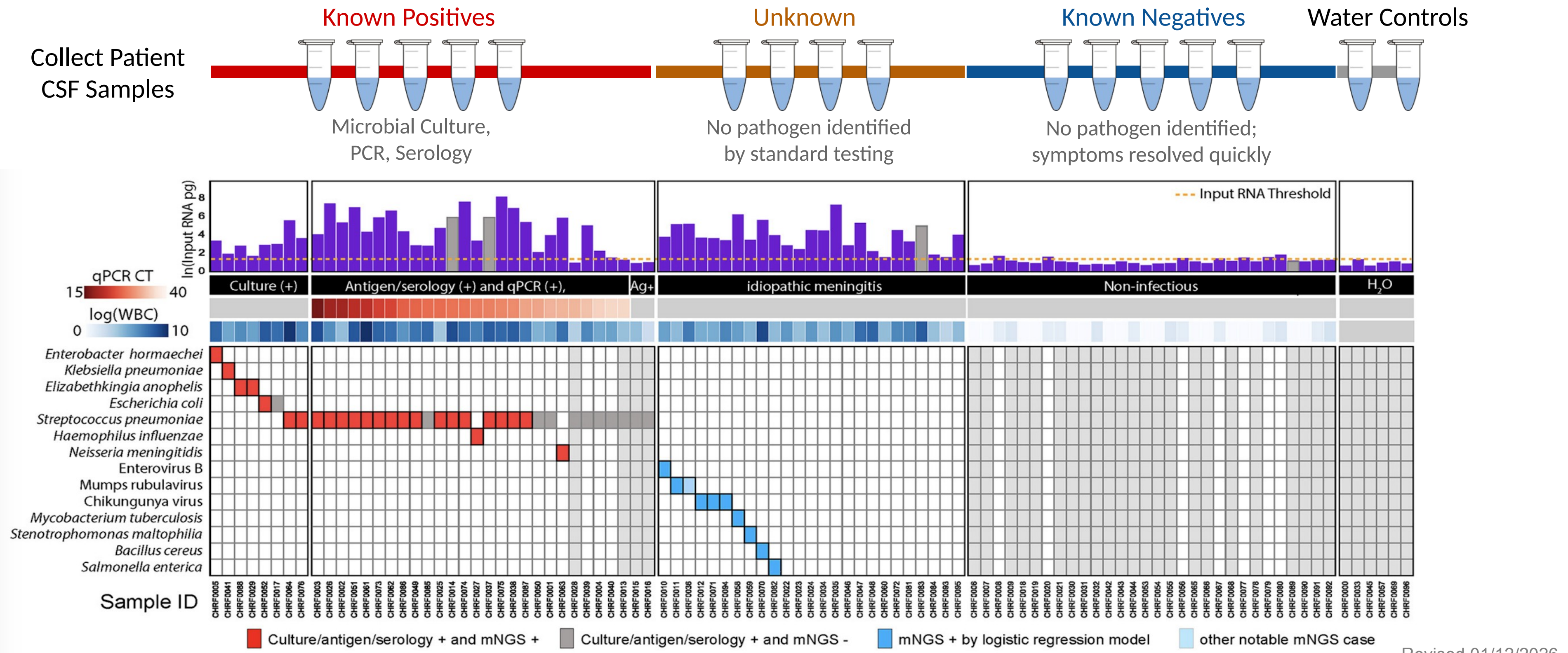


Characterize the microbial landscape



Use case: Saha et al., mBio, 2019

mNGS to identify etiologies of pediatric meningitis in Bangladesh

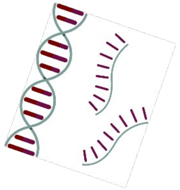


mNGS pipeline

Study design

- Type(s)
- Controls
- Storage
- Lab setup

xNA extraction
(1-2 hrs)



- Type(s)
- Controls
- QC

Library prep
(1-2 days)

- Host removal
- Fragmentation
- dsDNA generation
- End-repair
- Adapter ligation
- Barcoding & PCR
- Clean-up
- Pooling
- QC

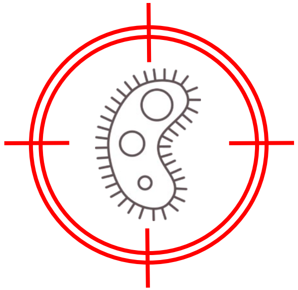
Sequencing
(0.5-3 days)

- Machine
- Read length
- Loading conc.

Data analysis
(0.5-4 hrs)

- QC data
- Host removal
- Alignment
- Blast
- Ranking of hits

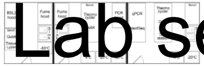
Pathogen ID



Revised 01/12/2026

Lab setup

*It is **<u>critical</u>** to have separate pre- and post-PCR lab spaces!*

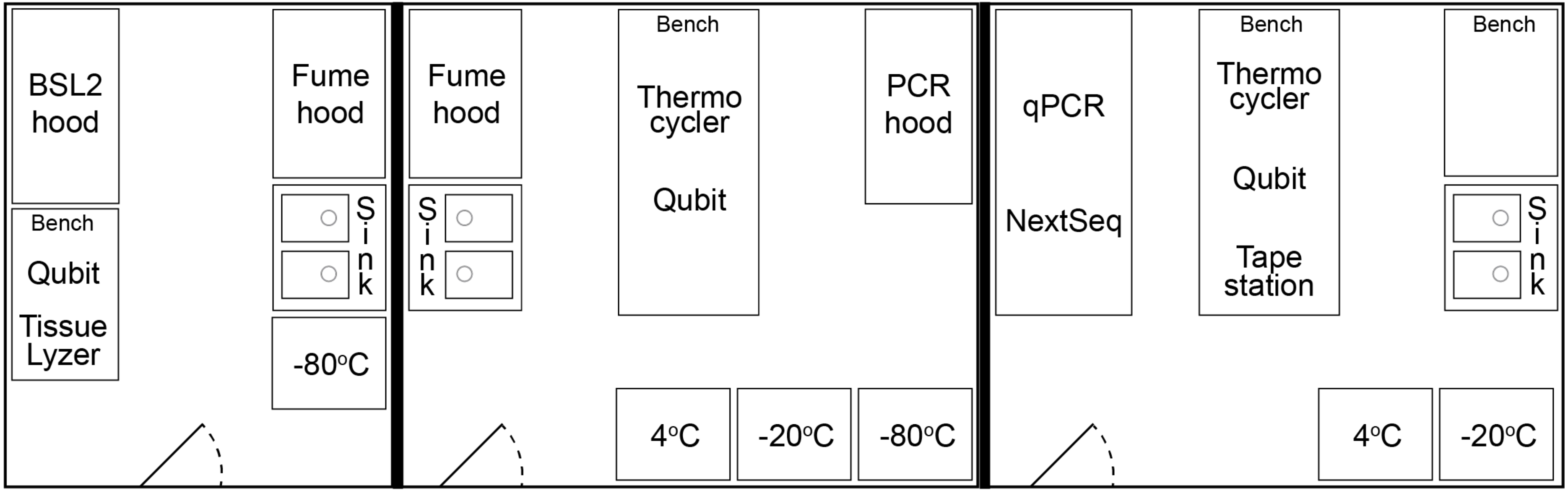
Lab setup

- Follow institutional guidelines for biosafety containment
  - Biosafety level 1 - 2 - … (BSL-1, BSL-2, …)
  - Note: Shield inactivates dangerous pathogens -> Biosafety level 1

- Avoid dust and high humidity levels

- Preventative measures to **avoid PCR amplicon contamination at all cost**
  - **Physical separation of Pre-PCR and Post-PCR**
  - Extra lab precautions (see next slides)

Lab setup



BSL-2 Pre-PCR

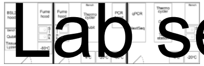
- **No PCR allowed**
- xNA extractions of infectious samples

BSL-1 Pre-PCR

- **No PCR allowed**
- xNA extractions
- Lib prep up to setting up the Q5 mastermix (no PCR)

BSL-1 Post-PCR

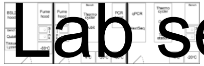
- **Lib prep from barcoding PCR**
- Lib QC
- Lib amplification if needed
- Sequencing



Lab setup

Extra lab precautions:

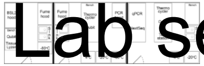
- Sticky floor pads at the entrance of Pre-PCR rooms, changed weekly
- Ice buckets color coded to the lab spaces - either Pre-PCR or Post-PCR only - no mixing
- Rotation of index primers so the same indices are never used in the same month
- Reagents like H_2O, SPRI beads, DNAse, Proteinase K, etc are aliquoted into single-use tubes
- Pipets inside the PCR hoods are never taken out. All hoods are sprayed down before and after use with 70% ethanol and RNAseZap
- Reagents are not shared between the different rooms
- PCR products never enter the Pre-PCR rooms nor samples with high nucleic acid quantities (i.e.: stool, tissues, bacterial cultures, bacterial colonies)



Lab setup

Other considerations:

- Power: Uninterruptible power supply (UPS) or surge protector
- Internet connection
- Environmental conditions, e.g. for iSeq:
  - Dust issues: Pollution degree rating of II (non-conductive pollution only, rare conductive pollution)
  - Ventilation: Up to 2048 BTU/hr @ 600W. Buy extra air filters for the iSeq.
  - Humidity: Non-condensing 20-80% Relative Humidity
  - Temperature: 15°C to 30°C (22.5°C ± 7.5°C)



Lab setup

Lab procedures by location

BSL2 Pre-PCR Biosafety Cabinet

- Sample inactivation in DNA/RNA Shield
- Nucleic acid extraction (if sample is not deactivated)

BLS2-Pre-PCR Room

- Storage of original samples at -80C

BSL1 Pre-PCR PCR hood

- Nucleic acid extraction (if sample is deactivated)
- Nucleic acid quantification/quality control:
    - Aliquots can be made in Pre-PCR and run in Post-PCR
- DNA/RNA library prep: Fragmentation to setting up the barcoding PCR

BSL1 Post-PCR PCR hood

- Post-PCR SPRI clean-up

BSL1 Post-PCR Bench

- Barcoding PCR amplification run
- Library quantification & quality control
- Library pooling
- qPCR Machine for additional library amplification
- Illumina sequencing
- Storage of NGS libraries -20C

Next: ***mNGS study design, lab setup and sample preparation***